

$$\varphi: A \longrightarrow A/\langle n \rangle$$

$$\ker \varphi = \langle n \rangle \Rightarrow \langle n \rangle = 0 \quad (\Rightarrow \in)$$

Prime ideals & Maximal Ideals

An ideal I of A is said to be prime ideal if $a, b \in A$
 $a, b \in I \Rightarrow$ either $a \in I$ or $b \in I$ &
 $I \subsetneq A$.

Ex: ① $p\mathbb{Z} \subseteq \mathbb{Z}$

$$(0) \subseteq \mathbb{Z}$$

Note: (0) is a prime ideal of $A \Leftrightarrow A$ is an integral domain.

② $\langle n \rangle \subseteq \mathbb{R}[x]$

$$\varphi: \mathbb{R}[x] \longrightarrow \mathbb{C}$$

$$f(x) \longmapsto f(i)$$

